457.557 Water Resources Systems Engineering, Spring 2022

16/30

Final Project A: A조 (김연주/이재황/이재호/박성열)

Due May-03, 2022

발표 시간 : 16분 38초 (-1)
Q+A : 2/5.

1. Create a DP optimization model for reservoir operation according to the conditions given in textbook 4.4.7.2 (p.120~137). Your model should be built in one of the following computer languages (R, Python, or Matlab NOT MICROSOFT EXCEL, although you may want to replicate first with Excel).

   You should write the codes by yourself (in teams-do not use already built DP model available online) and all the conditions for building the model can be found in the textbook. After running your model, your program should yield the results in Table 4.20 & Fig. 4.18.

   Prepare for a 15 min PPT presentation (around 28-35 slides) in Korean in class as teams. Answers for the following questions should be included during the presentation.

   4/5. A. How did you set up your model? Provide the key equations of your DP model (such as objective function, constraints, recursive equation, assumptions, etc.) and the ultimate objective of your model. Explain briefly regarding the conditions for optimality, convergence, etc. How is your solution's "optimality" defined?

   → Bellman optimality

   3/5. B. Briefly explain how the program you built was set (including the variables), and attach the codes at the end of the presentation.

   이수형 결과와 같이 $F^n - F^{n-4} = 186$ 을 보이는 형태임.
   → 새롭게 코드 작성 및 실행 필요.

   2/5. C. Do you think that the optimal solutions (the steady-state policy) is really optimal under all (or some) circumstances? Why or why not? Under which circumstances your optimal solutions hold value and do not hold value?

   → 이 부분에 대한 언급은?.

   2/5. D. Conduct a simulation of your model with Q1_t=24, 12, 6, 18 (for seasons t=1,2,3,4). Compare your results with the second case of Q2_t=6, 12, 24, 18. How does your model perform under the unexpected inflow scenario?

   이 부분에 대한 언급 X.

   4/5. E. Finally, present possible limitations of your model. How do you think that the limitations you presented can be overcome?